BEACONHOUSE NATIONAL UNIVERSITY

School of Computing & Information Technology

LAB ASSIGNMENT 5

Routing Protocol Redistribution

RIP <-> OSPF | Cisco Packet Tracer

Course	Computer Networks Lab (Lab-209)
Semester	Spring 2026
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Submission: .pkt file + screenshots + completed questions

1. Objective

The purpose of this lab is to configure two different routing protocol domains and connect them using route redistribution. The NayaTel side uses RIP, the PTCL side uses OSPF, and Router2 acts as the edge router where both protocols are redistributed.

2. IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Default Gateway / Connected To
PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
Router0	G0/0	192.168.1.1	255.255.255.0	Switch0 / LAN A
Router0	G0/1	10.0.1.1	255.255.255.0	Router1 G0/0
Router1	G0/0	10.0.1.2	255.255.255.0	Router0 G0/1
Router1	G0/1	10.0.2.1	255.255.255.0	Router2 G0/0
Router2	G0/0	10.0.2.2	255.255.255.0	RIP side / Router1 G0/1
Router2	G0/1	10.0.3.1	255.255.255.0	OSPF side / Router3 G0/0
Router3	G0/0	10.0.3.2	255.255.255.0	Router2 G0/1
Router3	G0/1	192.168.3.1	255.255.255.0	Switch1 / LAN B
PC2	NIC	192.168.3.10	255.255.255.0	192.168.3.1
Web Server	NIC	192.168.3.20	255.255.255.0	192.168.3.1
Router3	G0/2	10.0.4.1	255.255.255.0	Challenge link / Router4 G0/0
Router4	G0/0	10.0.4.2	255.255.255.0	Router3 G0/2
Router4	G0/1	192.168.4.1	255.255.255.0	PC3 gateway
PC3	NIC	192.168.4.10	255.255.255.0	192.168.4.1

3. Topology Screenshot

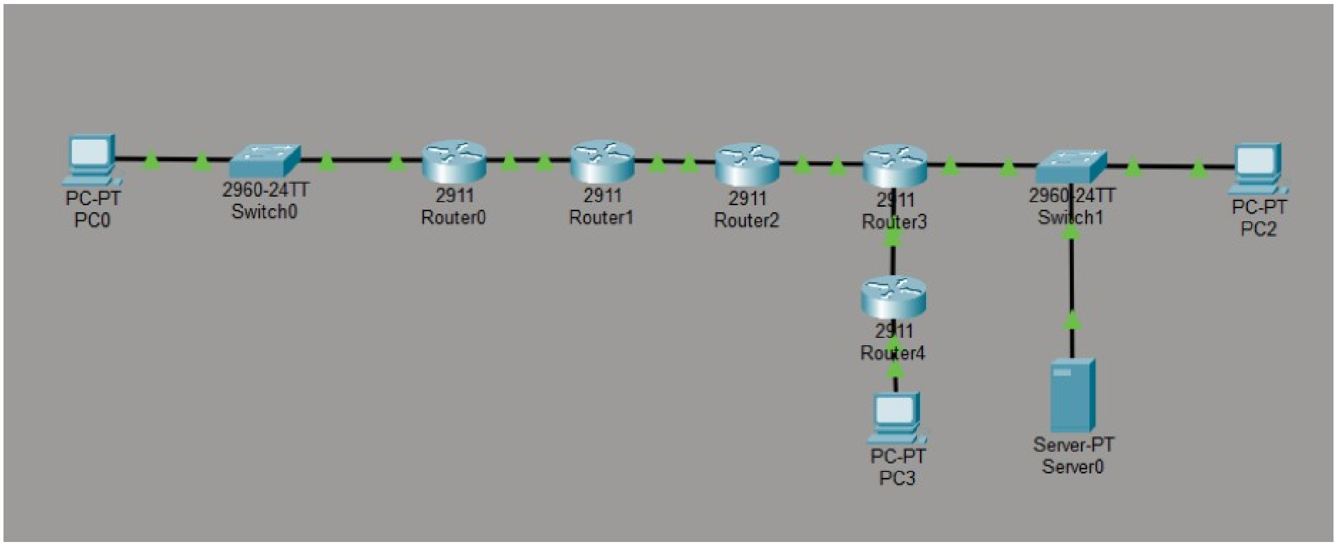

Figure 1: Final topology showing RIP domain, OSPF domain, Web Server, and bonus Router4/PC3 extension.

4. Router0 Routing Table Verification

```
Router0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.1.0/24 is directly connected, GigabitEthernet0/1
L       10.0.1.1/32 is directly connected, GigabitEthernet0/1
R       10.0.2.0/24 [120/1] via 10.0.1.2, 00:00:23, GigabitEthernet0/1
R       10.0.3.0/24 [120/2] via 10.0.1.2, 00:00:23, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.3.0/24 [120/11] via 10.0.1.2, 00:00:21, GigabitEthernet0/1
```

Figure 2: Router0 show ip route output showing redistributed OSPF-side networks as RIP routes.

5. Router3 Routing Table Verification

```
Router3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O E2    10.0.1.0/24 [110/20] via 10.0.3.1, 00:02:32, GigabitEthernet0/0
O E2    10.0.2.0/24 [110/20] via 10.0.3.1, 00:02:32, GigabitEthernet0/0
C       10.0.3.0/24 is directly connected, GigabitEthernet0/0
L       10.0.3.2/32 is directly connected, GigabitEthernet0/0
O E2 192.168.1.0/24 [110/20] via 10.0.3.1, 00:02:32, GigabitEthernet0/0
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
L       192.168.3.1/32 is directly connected, GigabitEthernet0/1
```

Figure 3: Router3 show ip route output showing RIP-side routes as OSPF external routes marked O E2.

6. Router2 OSPF Neighbor Verification

```
Router2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.3.1       1   FULL/BDR        00:00:30    10.0.3.2        GigabitEthernet0/1
```

Figure 4: Router2 show ip ospf neighbor output showing Router3 in FULL state.

7. PC0 Ping Test - Gateway

```
Command Prompt                                                              X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=7ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 1ms
```

Figure 5: PC0 ping to Router0 gateway 192.168.1.1 succeeded with 0% loss.

8. PC0 Ping Test - Router3

```
Pinging 10.0.3.2 with 32 bytes of data:

Reply from 10.0.3.2: bytes=32 time<1ms TTL=252
Reply from 10.0.3.2: bytes=32 time<1ms TTL=252
Reply from 10.0.3.2: bytes=32 time=10ms TTL=252
Reply from 10.0.3.2: bytes=32 time=1ms TTL=252

Ping statistics for 10.0.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Figure 6: PC0 ping to Router3 address 10.0.3.2 succeeded.

9. PC0 Ping Tests - End-to-End Connectivity

```
C:\>ping 10.0.2.2

Pinging 10.0.2.2 with 32 bytes of data:

Reply from 10.0.2.2: bytes=32 time<1ms TTL=253
Reply from 10.0.2.2: bytes=32 time=1ms TTL=253
Reply from 10.0.2.2: bytes=32 time<1ms TTL=253
Reply from 10.0.2.2: bytes=32 time=1ms TTL=253

Ping statistics for 10.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time<1ms TTL=124
Reply from 192.168.3.10: bytes=32 time=10ms TTL=124
Reply from 192.168.3.10: bytes=32 time=10ms TTL=124
Reply from 192.168.3.10: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 7ms

C:\>ping 192.168.3.20

Pinging 192.168.3.20 with 32 bytes of data:

Reply from 192.168.3.20: bytes=32 time<1ms TTL=124
Reply from 192.168.3.20: bytes=32 time<1ms TTL=124
Reply from 192.168.3.20: bytes=32 time=1ms TTL=124
Reply from 192.168.3.20: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.3.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 7: PC0 pings to 10.0.2.2, 192.168.3.10, and 192.168.3.20 succeeded with 0% loss.

10. PC2 Reverse Ping Test

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=124
Reply from 192.168.1.10: bytes=32 time<1ms TTL=124
Reply from 192.168.1.10: bytes=32 time<1ms TTL=124
Reply from 192.168.1.10: bytes=32 time=9ms TTL=124

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms
```

Figure 8: PC2 successfully pinged PC0 at 192.168.1.10.

11. PC0 Traceroute to PC2

```
C:\>tracert 192.168.3.10

Tracing route to 192.168.3.10 over a maximum of 30 hops:

  1   0 ms      0 ms      1 ms      192.168.1.1
  2   1 ms      0 ms      1 ms      10.0.1.2
  3   1 ms      0 ms      0 ms      10.0.2.2
  4   10 ms     0 ms      0 ms      10.0.3.2
  5   1 ms      0 ms      10 ms     192.168.3.10

Trace complete.
```

Figure 9: PC0 tracert to 192.168.3.10 shows the correct 5-hop path.

12. PC0 Web Browser Test to Web Server

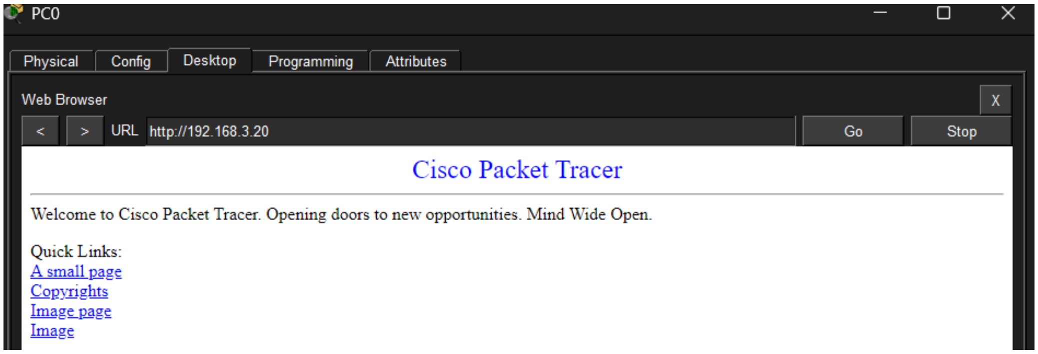

Figure 10: PC0 browser successfully opened the web server page at 192.168.3.20.

13. Challenge Task - Bonus Verification

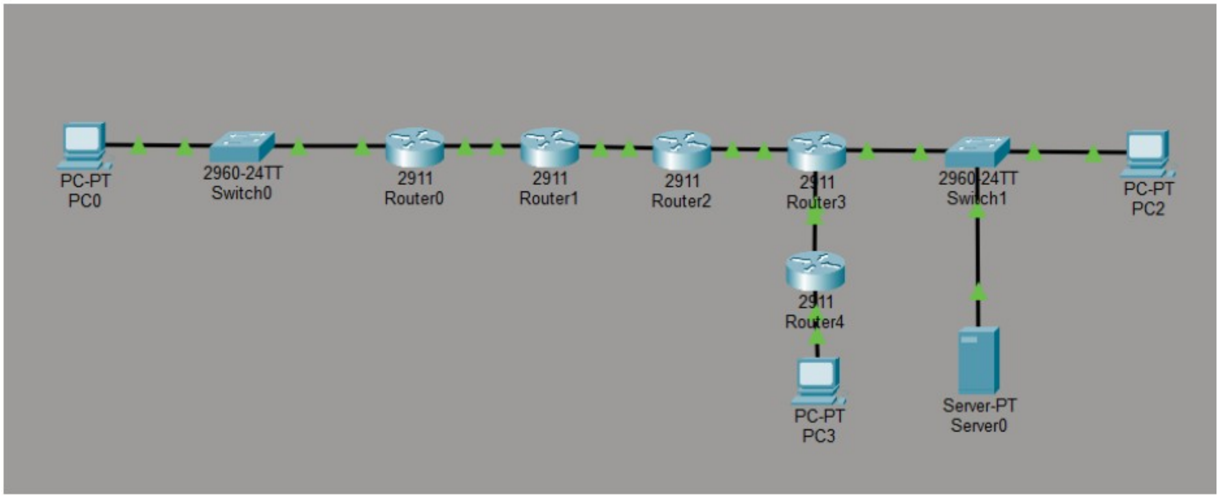

Figure 11: Challenge topology showing Router4 and PC3 added to the OSPF domain.

Router4 Interface Verification

```
Router4#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.4.2        YES manual up                    up
GigabitEthernet0/1     192.168.4.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Figure 12: Router4 interfaces G0/0 and G0/1 are configured and up/up.

Router4 OSPF Neighbor Verification

```
Router4#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.3.1       1   FULL/BDR        00:00:30    10.0.4.1        GigabitEthernet0/0
```

Figure 13: Router4 shows Router3 as an OSPF neighbor in FULL state.

PC0 Ping to PC3

```
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time<1ms TTL=123
Reply from 192.168.4.10: bytes=32 time=10ms TTL=123
Reply from 192.168.4.10: bytes=32 time=11ms TTL=123
Reply from 192.168.4.10: bytes=32 time<1ms TTL=123

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 5ms
```

Figure 14: PC0 successfully pings PC3 at 192.168.4.10 with 0% loss.

14. Main Configuration Commands Used

14.1 RIP Configuration

```
Router0(config)# router rip
Router0(config-router)# version 2
Router0(config-router)# network 192.168.1.0
Router0(config-router)# network 10.0.1.0
Router0(config-router)# no auto-summary

Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# network 10.0.1.0
Router1(config-router)# network 10.0.2.0
Router1(config-router)# no auto-summary

Router2(config)# router rip
Router2(config-router)# version 2
Router2(config-router)# network 10.0.2.0
Router2(config-router)# no auto-summary
```

14.2 OSPF Configuration

```
Router2(config)# router ospf 1
Router2(config-router)# network 10.0.3.0 0.0.0.255 area 0

Router3(config)# router ospf 1
Router3(config-router)# network 10.0.3.0 0.0.0.255 area 0
Router3(config-router)# network 192.168.3.0 0.0.0.255 area 0
```

14.3 Route Redistribution on Router2

```
Router2(config)# router ospf 1
Router2(config-router)# redistribute rip subnets
Router2(config-router)# exit

Router2(config)# router rip
Router2(config-router)# redistribute ospf 1 metric 10
```

14.4 Challenge Commands

```
Router3(config)# interface gigabitEthernet0/2
Router3(config-if)# ip address 10.0.4.1 255.255.255.0
Router3(config-if)# no shutdown
Router3(config)# router ospf 1
Router3(config-router)# network 10.0.4.0 0.0.0.255 area 0

Router4(config)# interface gigabitEthernet0/0
Router4(config-if)# ip address 10.0.4.2 255.255.255.0
Router4(config-if)# no shutdown
Router4(config)# interface gigabitEthernet0/1
Router4(config-if)# ip address 192.168.4.1 255.255.255.0
Router4(config-if)# no shutdown
Router4(config)# router ospf 1
```

```
Router4(config-router)# network 10.0.4.0 0.0.0.255 area 0
Router4(config-router)# network 192.168.4.0 0.0.0.255 area 0
```

15. Completed Task Questions

Task 1: What is the IP address of Router2's G0/1 interface?

Answer: 10.0.3.1

Task 1: What two domains does Router2 connect?

Answer: NayaTel/RIP domain and PTCL/OSPF domain.

Task 2: What letter appears next to RIP-learned routes in the routing table?

Answer: R

Task 2: What is the administrative distance of RIP?

Answer: 120

Task 2: Why should 10.0.3.0 NOT be added to RIP on Router2?

Answer: Because 10.0.3.0 belongs to the OSPF side. Router2 should learn and share that network through OSPF and redistribution, not by directly advertising it in RIP.

Task 3: What letter appears next to OSPF-learned routes?

Answer: O

Task 3: What is the administrative distance of OSPF?

Answer: 110

Task 3: What OSPF neighbour state indicates a fully working adjacency?

Answer: FULL

Task 3: Compare OSPF and RIP: which has lower administrative distance, and why does it matter?

Answer: OSPF has the lower administrative distance, 110, while RIP has 120. A lower administrative distance means the router trusts that route more if multiple routing protocols provide routes to the same destination.

Task 4: After redistribution, what route marker appears on Router3 for routes learned from the RIP domain?

Answer: O E2

Task 4: What does the subnets keyword do in redistribute rip subnets?

Answer: It allows classless subnet routes to be redistributed into OSPF, not only classful networks.

Task 4: Why is metric 10 needed when redistributing OSPF routes into RIP?

Answer: RIP uses hop count as its metric, so a metric must be assigned to OSPF routes when they are injected into RIP.

Task 5: Did all 5 ping tests from PC0 succeed?

Answer: Yes.

Task 5: How many hops did the traceroute from PC0 to PC2 show?

Answer: 5 hops.

Task 5: Which router represents the redistribution boundary?

Answer: Router2. Boundary IPs: 10.0.2.2 and 10.0.3.1.

Task 5: Did PC2 successfully ping PC0 in reverse?

Answer: Yes.

16. Challenge Questions

What OSPF commands did you add to Router4?

Answer: network 10.0.4.0 0.0.0.255 area 0
network 192.168.4.0 0.0.0.255 area 0

After adding Router4, did PC0 successfully ping PC3 (192.168.4.10)?

Answer: Yes.

Did you need to change anything on Router2?

Answer: No.

Why or why not?

Answer: No changes were required on Router2 because redistribution was already configured. Router4's new network was added into OSPF, and Router2 automatically redistributed the new OSPF route into RIP.

17. Conclusion

The lab was completed successfully. RIP was configured on the NayaTel side, OSPF was configured on the PTCL side, and Router2 was used as the redistribution boundary. Routing tables, OSPF neighbor status, ping tests, traceroute, reverse ping, web browser testing, and the bonus Router4/PC3 ping test confirm that end-to-end connectivity is working across both routing protocol domains.